

Linear Regression

A Comprehensive Overview

1. What is Linear Regression?

Linear Regression is one of the most fundamental and widely used algorithms in statistics and machine learning. It models the relationship between one or more **independent variables (features)** and a **dependent variable (target)** by fitting a straight line (or hyperplane) through the data. The goal is to find the best-fit line that minimizes the difference between predicted and actual values.

There are two main types:

- **Simple Linear Regression** — one independent variable (X) predicts one dependent variable (Y).
- **Multiple Linear Regression** — two or more independent variables predict one dependent variable.

2. The Linear Regression Equation

The equation of a simple linear regression line is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Symbol	Name	Description
Y	Dependent Variable	The outcome we want to predict
X	Independent Variable	The input feature / predictor
β_0	Intercept (bias)	Value of Y when X = 0
β_1	Slope (coefficient)	Change in Y per unit change in X
ϵ	Error Term	Residual / unexplained variation

3. How Does It Work?

The algorithm learns the values of β_0 and β_1 that best fit the data by minimizing the **Sum of Squared Errors (SSE)**, also called the **Ordinary Least Squares (OLS)** method.

Cost Function (Mean Squared Error):

$$MSE = (1/n) \times \sum (Y_i - \hat{Y}_i)^2$$

Where Y_i is the actual value and $\hat{Y}_i$ is the predicted value. The model adjusts the slope and intercept iteratively (via calculus or gradient descent) until this error is minimized.

Linear Regression — Continued

4. Key Assumptions

- ✓ **Linearity:** The relationship between X and Y is linear.
- ✓ **Independence:** Observations are independent of each other.
- ✓ **Homoscedasticity:** Constant variance of residuals across all levels of X.
- ✓ **Normality:** Residuals (errors) are normally distributed.
- ✓ **No Multicollinearity:** In multiple regression, predictors are not highly correlated with each other.

5. Model Evaluation Metrics

Metric	Formula	Interpretation
R ² Score	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$	Proportion of variance explained (0–1; higher = better)
MSE	$\text{mean}((Y - \hat{Y})^2)$	Average squared error; penalizes large errors
RMSE	$\sqrt{\text{MSE}}$	Same unit as Y; easy to interpret
MAE	$\text{mean}(Y - \hat{Y})$	Average absolute error; robust to outliers

6. Advantages & Limitations

✓ Advantages	✗ Limitations
Simple and interpretable	Assumes a linear relationship
Computationally efficient	Sensitive to outliers
Works well on linearly separable data	Cannot model complex patterns
Provides statistical significance of features	Affected by multicollinearity
Good baseline model	Requires feature scaling in some cases

7. Real-World Applications

- **House Price Prediction:** Estimate property value based on size, location, and age.
- **Sales Forecasting:** Predict future revenue from advertising spend.
- **Medical Research:** Relate dosage to patient response or blood pressure to BMI.
- **Finance:** Predict stock returns or credit risk scores.
- **Agriculture:** Estimate crop yield based on rainfall and temperature.

Linear Regression remains a cornerstone of predictive modeling — simple, powerful, and interpretable.